

FINANCIAL FORECASTING: STOCK PRICE PREDICTION

TRANSFORMER BASED DEEP LEARNING
MODEL FOR DAILY STOCK RETURN
PREDICTION: A COMPARATIVE STUDY WITH
RECURRENT NEURAL NETWORK

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INTRODUCTION

Investment in financial market has been one of the common form of investments around the globe for a long time. People who invests in it always try to gain a maximum profit. Financial markets are interesting as well as volatile in nature, which makes investment quite challenging. The study of stock price prediction is a way to understand how the market functions. To study the financial forecasting, the most important part is the data, which is generated in large volumes and it makes it a suitable domain of study for data science and machine learning research. There are various ways to predict the stock price. There are traditional statistical approaches such as the **Autoregressive Integrated Moving Average (ARIMA)** which has been widely used in financial forecasting. However, there are certain drawbacks in this model as they rely heavily on linear relationships and fail to understand the complex nonlinear market behaviour which are present in the financial time-series data. They also fail to incorporate multiple features.

With the recent advancement of studies in areas of machine learning and deep learning methods, they have been widely used in financial prediction tasks. Models such as **Recurrent Neural Network (RNN)** and **Long Short-Term Memory (LSTM)** have given improved results as compared to traditional statistical approach. The main reason for this improvement is the ability to process time-series data sequentially by remembering the past information. In spite of the given advantages, these models have certain limitations such as vanishing gradients, difficulty modelling long-term dependencies within large sequences and slow training due to sequential training.

The **Transformer model** was first introduced in the paper Attention Is All You Need (Vaswani et al., 2017). This model has really helped improve sequence modelling. Transformers use a mechanism called self-attention. This self-attention helps the model pay attention to parts of the input sequence. It does this without needing to process the sequence step by step like models did. The Transformer model is based on self-attention mechanism, which makes it very good at handling sequences and it has made a difference in sequence modelling. This capability enables the model of long-range dependencies, parallel computation, and feature interaction which improves computational efficiency.

The working of the Transformer model is unique and is called the self-attention mechanism. This mechanism identifies the importance of different elements within the data which has been inputted. This working mechanism is highly helpful with financial time-series data, as the model learns the importance of the past observations to predict the future returns on stock.

There are several features of the Transformer model. One is that it consist of an encoder and decoder. The encoder helps in processing the input data and generates an output of representations

of each element in the sequence. Another important feature of this model is positional encoding. This helps the model to provide information in which order the elements are in the input data.

The Transformer model has several advantages compared to recurrent neural network models. The Transformer model allow parallel processing which makes it more efficient compared to machine learning and deep learning methods.

In this research, a comparative study will be done between the Transformer model, the recurrent neural network and the traditional statistical models to understand how each model works, their advantages and disadvantages and how Transformer model is better than the others. It will also discuss the shortcomings of the Transformer model. For the prediction of the daily stock returns, historical data will be used and the performance of each model will be evaluated.

The research will also study if there is an improvement in the prediction of daily stock returns in the Transformer model compared to the other traditional statistical models and recurrent neural network models.

AIM AND OBJECTIVES

The aim of the research is to develop a Transformer model for daily stock return prediction and to evaluate it with recurrent neural network models and other traditional statistical models. The research will start with the traditional statistical method and then work through other machine learning methods and will finally develop the Transformer model and address the advantages as well as the shortcomings of the models.

OBJECTIVES

1. To collect historical data preferably for 15 years from reliable financial data sources.
2. To perform exploratory data analysis on the given data and calculate daily stock returns and the direction of how the stocks will behave.
3. To start prediction using traditional statistical approaches and then moving onto deep learning methods such as Recurrent Neural Network. To understand how they work and to note the drawbacks as well
4. To develop a Transformer based deep learning model for stock price prediction and to understand its working and how it addresses the drawbacks of the other models which has been analysed.
5. To develop a comparative study of the models and evaluate the performance of Transformer and the other models using evaluation methods.

6. To analyse the strengths and limitations of Transformer models in financial time-series forecasting.

LITERATURE REVIEW

Stock price prediction has been a very popular subject of study as they are best methods of investments but they also carry a certain level of risk due to its volatility. Due to its popularity, various methods were used for price prediction. The traditional statistical models such as ARIMA have been used widely due to its simplicity and interpretation (Fama, 1970). This model had its drawback as it assumed linear relationship of the data provided and failed to interpret the complex nature of financial market.

To overcome these limitations, various machine learning methods such as Random Forest models were developed. These models took into account the market sentiment for the prediction. In spite of the fact that they were able to successfully take into account the non-linear relationship but they relied on the manual feature engineering (Gu, Kelly and Xiu, 2020).

Deep learning models such as Recurrent Neural Networks, were better as they had the capability to automatically learn the complex features from financial raw data. Hence such models were widely used for financial time-series forecasting as the data was processed sequentially (Hochreiter and Schmidhuber, 1997). Despite the advantages, these deep learning models also had limitations. Training these models proved to be difficult. Another drawback was their limited ability to capture dependencies in long-term which reduced their performance in financial data (Sezer, Gudelek and Ozbayoglu, 2020).

Transformer models seemed to address the drawbacks of the other models as it worked on the attention mechanism (Vaswani et al., 2017). It works by processing the data in parallel and not in a sequence.

The research will evaluate the models and try to conclude which model works better. It will be done by testing the accuracy of the models and make a comparative study and simulate investment strategies.

PROPOSED RESEARCH DESIGN:

The research will experiment with the various designs starting with the traditional statistical model then working through the various deep learning models and ultimately concluding with Transformer

model to infer as to which model is ideal for stock price prediction and also discuss the various limitations each have.

The various steps that are to be followed are:

Data Collection:

To start with the research, the most important step is the data collection. For the research, historical data are to be obtained from reliable financial source such as Yahoo Finance and the Bloomberg terminal which is available at the University. Apart from the mentioned sources, another reliable data source is the Alpha Vantage website. These data are publicly available and have API which makes it easy to process in the programming interface. The dataset will include the daily financial data such as:

- Open Price
- High Price
- Low Price
- Close Price
- Adjusted close price
- Trading Volume

The financial data will be collected for multiple stocks over the period of 15 years for better understanding of how the stock price behaves. The idea is to include the 2008 data to understand the global crisis as well as the 2020 data to analyse the COVID crash.

Exploratory Data Analysis

The historical data after being obtained has to be pre-processed before building a model. It needs to be explored and seen if any values are missing or if any input feature needs to be standardised. After this the calculation is to be done for daily stock return. This research will primarily focus on prediction to be made on the stock return and the direction in which a stock might behave that is either it will move up or down. The prediction of stock price will not be done as they are not stationary.

Model Development

The research will start with the development of the simple statistical model such as Autoregressive Integrated Moving Average. It will explore how it works and also understand its shortcomings. Next it will move to deep learning models such as Recurrent Neural Network and will ultimately implement the Transformer based model. There will be a comparative study of these models and

inference will be made on difference between each of them, the shortcomings of the models and how the Transformer architecture is better than the others.

All the models will be implemented using Python libraries such as TensorFlow and PyTorch for learning.

Evaluation Metrics:

The performance of the model will be evaluated using metrics that are commonly used for prediction of stock prices such as Mean Squared Error and Root Mean Squared Error. During the research project other metrics might be added as per the requirement. The results will give a comparative study between the various models and also help determine why Transformer model is better than the other models.

PROJECT RISK ASSESSMENT

Every research project has its own risk and the risk with stock price prediction is quite high due to the volatility of the market.

One of the risk is the limited access to the financial datasets. To mitigate such risk, three sources have fixed: Bloomberg terminal which the University has access, Yahoo Finance where historical data can be downloaded free and Alpha Vantage which has API for it to be accessed by programming interface.

Another potential risk is the computational requirements. Training models especially Transformer architectures, may require significant resources which only professional settings are able to perform. Hence to minimise this risk, cloud based platform such as Google Collab and University computing facilities will be utilised.

Another major risk is the timeline for completion of the project. The research is analysing the historical stock price over the period of 15 years. The number of stocks that are to be analysed is a minimum of 30. For this a structured project timeline and planning is needed to complete the project successfully.

ETHICS

The proposed research will use historical data that are available publicly. This will not involve any personal or sensitive information. Hence the project is likely to raise any ethical concern as it will not have any human participant.

The research will be in compliance to the University Ethical Policy. Since it uses publicly available secondary financial data, it is unlikely that any ethical approval will be needed.

All the data sources and the academic literature will be cited appropriately to maintain integrity and avoid plagiarism.

IMPACT

The research is based on artificial intelligence applications in financial forecasting and has the potential to contribute in this ever growing field. In this research a comparative study will be conducted on the various models, starting from the traditional statistical model, deep learning model and Transformer architecture. In this way a proper evaluation can be conducted to infer the effectiveness of the Transformer-based model for stock price prediction.

The comparative study will help to understand the various nuances of the models and might contribute to the development of a more accurate financial model which will help to make accurate readings. These readings could support investment decision and also manage financial risk management.

If seen from a broader perspective, the improvement in forecasting models might support for efficient financial markets by helping in better data-driven decision-making.

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